# Sixun Dong

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## Research Profile

I am a Ph.D. student at the Institute of Artificial Intelligence, University of Central Florida, advised by Dr. Chen Chen. I build multimodal AI systems that perceive, generate, and act. My work began in video perception and representation learning (CVPR'22 Oral, CVPR'23), extended to generative modeling (3DV'24, NeurIPS'25), and now follows three threads: **efficient multimodal and long-sequence inference** (ICLR'26, NeurIPS'26 ×2); **long-horizon LLM agents** (ECCV'26, ICASSP'26); and **evaluating and improving LLM capabilities**, i.e., finding where metrics fail and training models against corrected signals (COLM'26, WACV'25). Going forward, I study system–algorithm co-design for agent harnesses: the inference, memory, and orchestration layers behind long-horizon multimodal agents. Current projects span this stack: orchestration (long-horizon game agents), long-video memory, deciding when to act (proactive streaming assistants), and efficient reasoning.

## Education

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| 2026 – Present | **Ph.D. student in Computer Science**, University of Central Florida, USA | Advisor: Dr. Chen Chen |
| 2024 – 2026 | **Ph.D. student in Computer Science**, Arizona State University, USA | |
| 2021 – 2024 | **M.S. in Computer Science**, ShanghaiTech University, China | Advisor: Prof. Shenghua Gao |
| 2017 – 2020 | **B.E. in Computer Science** (dual degree), Dalian University of Technology, China | |
| 2016 – 2020 | **B.E. in Process Equipment & Control Engineering**, Dalian University of Technology, China | |

## Research Experience

May–Aug 2026 **Tencent IEG**, Motus AI Team — Shanghai, China
*Research Intern, Tencent Qingyun Program*
- Built **MotionFact**, a typed rubric of fine-grained motion facts that exposes blind spots of similarity- and LLM-based captioning metrics, plus **MotionADV**, an adversarial benchmark; a decomposed judge (**MotionJudge**) reaches $\geq$0.90 detection and $\geq$0.89 pass rates on every physical attribute.
- Developed **Motion-VLM**, a multi-view VLM with frame-aligned 3D motion tokens, outperforming five published motion captioners on HumanML3D. First-author paper [7], under review at ICLR 2027.

May–Aug 2025 **Zoom Communications**, GenAI Research — Seattle, WA, USA
*Research Intern*
- Proposed **MMTok**, training-free multimodal token selection for efficient VLM inference, retaining 87.7% performance with only 4 vision tokens [3] (first author, ICLR'26).
- Contributed to **LogicIF** [9] (complex logical instruction following, COLM'26) and **LiveMCP-101** [11] (MCP agent stress-testing).

2023 – 2024 **DGene Digital Technology**, Digital Human Algorithm Dept. — Shanghai, China
*Research Intern*
- Led three digital-human projects: audio-driven talking heads with facial restoration (outperformed industrial and academic SoTA); multi-view RGB to parametric 3D body reconstruction (<7% anthropometric error); co-speech gesture generation aligned with head pose.

## Selected Publications

★ first / co-first author  * equal contribution  First-author work first within each theme.  Full list on Google Scholar

### *Efficient multimodal & long-sequence inference*

[1] NeurIPS'26 ★ **Rethinking Long-Video Efficiency: A Joint Allocation Perspective on Frames, Pixels, and Front-End Latency.**
**Sixun Dong**, Wei Li, Andong Deng, Qi Qian, Victor Zhu, Zhengping Ji, Chen Chen. [Paper] [OpenReview] [Project]
*— with Axon & Meta*

[2] *arXiv'26* ★ **Rethinking Model Efficiency: Multi-Agent Inference with Large Models.**
**Sixun Dong**, Juhua Hu, Steven Li, Wei Wen, Qi Qian. [Paper]
*— with Meta & UW*

[3] ICLR'26 ★ **MMTok: Multimodal Coverage Maximization for Efficient Inference of VLMs.**
**Sixun Dong**, Juhua Hu, Mian Zhang, Ming Yin, Yanjie Fu, Qi Qian. [Paper] [Code]
*— Zoom internship*

[4] NeurIPS'26 **MARS: Multi-resolution Adaptive Routing for Sequential Recommendation.**
Ming Yin, **Sixun Dong**, Yudong Liu, Wen-Yun Yang, Yunjiang Jiang, Yiran Chen. [Paper] [OpenReview]
*— with Duke & Meta*

### *Long-horizon agents: orchestration, memory, execution*

[5] ECCV'26 ★ **Better Call CineCrew: Consistent Ultra-Long Narrative-to-Film Generation.**
Jiaben Chen*, **Sixun Dong***, Qinhong Zhou, Raine Ma, Zhiyang Dou, Wojciech Matusik, Chuang Gan. [Paper] [Code] — *with UMass & MIT*

[6] ICASSP'26 **Towards Robust Dysarthric Speech Recognition: LLM-Agent Post-ASR Correction Beyond WER.**
Xiuwen Zheng, **Sixun Dong**, Bornali Phukon, Mark Hasegawa-Johnson, Chang D. Yoo. [Paper] [Code] — *with UIUC*

### *Evaluating & improving LLM capabilities*

[7] *Under Review* ★ **Weak Models or Insensitive Metrics? Auditing and Training Fine-Grained Motion Captioning.**
**Sixun Dong**, Sheng Liu, Yiqing Xu, Jinyi Wang, Xinwei Jiang, Chen Chen, Zijiao Zeng. — *Tencent internship*

[8] *Under Review* ★ **Sensitivity or Criterion? Measuring and Moving When Language Models Act.**
**Sixun Dong**, Yebowen Hu, Ming Yin, Yiran Chen, Fei Liu, Chen Chen. [Paper] [Project] — *with Emory & Duke*

[9] COLM'26 **LogicIF: Towards Complex Logic Instruction Following.**
Mian Zhang, Shujian Liu, **Sixun Dong**, Ming Yin, Yebowen Hu, Xun Wang, Steven Ma, Song Wang, Sathish Reddy Indurthi, Haoyun Deng, Zhiyu Zoey Chen, Kaiqiang Song. [Paper] [Code] — *Zoom internship*

[10] WACV'25 **MLLM-Tool: A Multimodal Large Language Model for Tool Agent Learning.**
Chenyu Wang, Weixin Luo, **Sixun Dong**, Xiaohua Xuan, Zhengxin Li, Lin Ma, Shenghua Gao. [Paper] [Code]

[11] *arXiv'25* **LiveMCP-101: Stress Testing and Diagnosing MCP-enabled Agents on Challenging Queries.**
Ming Yin, Dinghan Shen, Silei Xu, Jianbing Han, **Sixun Dong**, Mian Zhang, Yebowen Hu, Shujian Liu, Simin Ma, Song Wang, Sathish Reddy Indurthi, Xun Wang, Yiran Chen, Kaiqiang Song. [Paper] — *Zoom internship*

### *Perception & generative modeling*

[12] CVPR'23 ★ **Weakly Supervised Video Representation Learning with Unaligned Text for Sequential Videos.**
**Sixun Dong***, Huazhang Hu*, Dongze Lian, Weixin Luo, Yicheng Qian, Shenghua Gao. [Paper] [Code]

[13] CVPR'22 Oral ★ 🏆 **TransRAC: Encoding Multi-scale Temporal Correlation with Transformers for Repetitive Action Counting.**
Huazhang Hu*, **Sixun Dong***, Yiqun Zhao, Dongze Lian, Zhengxin Li, Shenghua Gao. [Paper] [Code]

[14] *arXiv'25* ★ **Teaching Time Series to See and Speak: Forecasting with Aligned Visual and Textual Perspectives.**
**Sixun Dong**, Wei Fan, Teresa Wu, Yanjie Fu. [Paper] [Code]

[15] NeurIPS'25 **Sculpting Features from Noise: Reward-Guided Hierarchical Diffusion for Task-Optimal Feature Transformation.**
Nanxu Gong*, Zijun Li*, **Sixun Dong**, Haoyue Bai, Wangyang Ying, Xinyuan Wang, Yanjie Fu. [Paper] [Code]

[16] 3DV'24 **RoomDesigner: Encoding Anchor-latents for Style-consistent and Shape-compatible Indoor Scene Generation.**
Yiqun Zhao, Zibo Zhao, Jing Li, **Sixun Dong**, Shenghua Gao. [Paper] [Code]

## Honors & Awards

2026 **Tencent Qingyun Program**, Tencent's top young-talent research initiative
2026 **ICLR Financial Assistance Award**
2023 **CVPR Travel Award**
2022 🏆 **CVPR Oral Presentation** (top 4% of submissions)

## Academic Service

Conference Reviewer for CVPR (2023–2026), ICCV (2023, 2025), ECCV (2024, 2026), NeurIPS (2025), ICML (2025, 2026), ICLR (2026), ACM MM (2023–2025), ACCV (2024), KDD (2025)
Journal Reviewer for IEEE Transactions on Multimedia, Neural Networks, ACM TKDD